Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines retrieval with language generation. Instead of relying only on what a language model memorised during training, a RAG system first retrieves relevant passages from an external knowledge source and then uses them as context for the answer. This reduces hallucinations and allows answers to be grounded in up-to-date or private documents.

The retriever finds relevant document passages. Documents are split into chunks, each chunk is converted into an embedding, and the embeddings are stored in a vector database such as FAISS or Chroma. When a question arrives, it is embedded and the database returns the top-k most similar chunks.

Vector Databases

A vector database stores embedding vectors together with metadata such as the chunk text, the source file name and a chunk id. Similarity search compares the query vector with stored vectors using cosine similarity or inner product, and returns the nearest neighbours. FAISS is a library from Meta for efficient similarity search over dense vectors.

Chunking

Chunking splits long documents into smaller overlapping pieces. Overlap keeps context that would otherwise be lost at chunk boundaries. Good chunk size balances precision, because small chunks are more specific, against context, because large chunks carry more surrounding information.